



Change Management – Lecture 10

What is Change Management?



- A structured process to manage modifications to systems, applications, and configurations.



- Ensures changes are planned, tested, reviewed, approved, and documented.



- Aims to minimise disruption, maintain security posture, and prevent misconfigurations.

Objectives of Change Management

Preserve

- Preserve availability, security, and integrity.

Prevent

- Prevent unauthorised changes.

Reduce

- Reduce downtime and change-related incidents.

Comply

- Comply with audit and regulatory requirements.

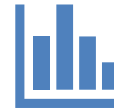
Key Components of Change Management



- Stakeholder Input



- Change Advisory Board (CAB)



- Impact Analysis



- Test and Rollout Plans



- Backout Plans



- Standard Operating Procedures (SOPs)



- Maintenance Windows

Change Types and Governance

- Standard Change: Pre-approved, low-risk
- Normal Change: Goes through full CAB review
- Emergency Change: Requires immediate implementation, post-review required
- Aligned with policies, procedures, and governance structures

Restarts, Dependencies, and Downtime

- Some changes require system reboots or may affect dependent services
- Managing legacy systems is often more complex
- Change planning includes:
 - Downtime scheduling
 - Stakeholder notifications
 - Business continuity considerations

Documentation and Version Control

- - Every change should update:
 - • System documentation
 - • Network diagrams
 - • Operational manuals
- - Use version control tools to:
 - • Track changes
 - • Maintain audit trails
 - • Support rollback if necessary

Security Role in Change Management

Security team reviews for:

Compliance with security standards

Risk assessment

Enforcement of least privilege and
change logging

Integrate with SIEM, logging, and
alerting systems

Automation & Orchestration

- Automation: Reduces human error, speeds up change implementation
- Orchestration: Coordinates changes across multiple systems
- Helps standardise changes and maintain consistency
- Supports auditability and compliance

Summary

- Change Management is essential for secure, stable, and compliant IT operations

- Aligns with security, business continuity, and regulatory goals

- Documentation, communication, and structured approvals are key

- Automation enhances efficiency but must be governed